



Unexpected curves arising from special line arrangements

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Abstract

In a recent paper, Cook et al. (Compos Math 154:2150–2194, 2018) used the splitting type of a line arrangement in the projective plane to study the number of conditions imposed by a general fat point of multiplicity j on the linear system of curves of degree $j + 1$ passing through the configuration of points dual to the given arrangement. If the number of conditions is less than the expected, they said that the configuration of points admits unexpected curves. In this paper, we characterize supersolvable line arrangements whose dual configuration admits unexpected curves and we provide other infinite families of line arrangements with this property.

Keywords Fat points · Line arrangements · Linear systems · Splitting types

Mathematics Subject Classification 14N20 (primary) · 13D02 · 14C20 · 14N05 · 05E40 · 14F05 (secondary)

1 Introduction

Polynomial interpolation problems are among the most studied topics in algebraic geometry. A classical example deals with computing dimensions of linear systems of curves of given degree passing through a given set of points in the projective plane. In other words, on the space of coefficients of ternary homogeneous polynomials of

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degree j , we consider the system of linear equations given by imposing the vanishing at a set of d points and we want to study the dimension of its solution. If the points are in *general position*, we may assume that this system of linear equations has maximal rank and the dimension of the solution is as small as possible, i.e., it is equal to $\binom{j+2}{2} - d$, unless this difference is negative, in which case it is zero [11].

If we consider points with some *multiplicity*, usually called *fat points*, where we require that the partial derivatives of the polynomials up to some order vanish at the points, the problem becomes much more complicated and only poorly understood. In other words, we consider a polynomial interpolation problem where we look at plane curves having singularities of certain order at a set of points. A complete answer is not known, even for points in *general position*.

Here is an example where the solution is not as expected. Consider the space of plane quartics, which has dimension $\binom{4+2}{2} = 15$, and consider a scheme of five double points in general position, i.e., we consider the linear system of plane quartics having five singularities at general points. Imposing a singularity at a point provides three linear equations, i.e., the vanishing of the three partial derivatives. Therefore, we have a system of 15 linear equations on the space of plane quartics and we expect to have no quartics with five general singularities. However, through five general points there exists always a conic and, therefore, the double conic is an unexpected quartic singular at every point and, in particular, at the set of five general points.

The case up to nine general points goes back to Castelnuovo, and it can also be found in the work of Nagata [20]. In the 1980s, Harbourne [17], Gimigliano [12] and Hirschowitz [19] independently gave conjectures on the dimension of a linear system of plane curves of given degree and with multiple general base points. In [3], these conjectures have been proved to be all equivalent to an older conjecture by Segre [23] and, for this reason, we refer to them as the *SHGH conjecture*.

In a recent paper, Cook et al. [4] slightly changed the question. Instead of counting the number of linear conditions given by a set of general multiple points on the complete linear system on plane curves of given degree, as in the classical problem, they look at the conditions imposed by a general fat point to the linear system of plane curves of given degree and passing through some particular configuration of reduced points.

This new question was motivated by previous works. Faenzi and Vallès noticed the relation between the *splitting type* of a line arrangement and curves passing through the set of points dual to the line arrangement and a fat point of multiplicity one less than the degree of the curve [10]. Afterward, Di Gennaro, Ilardi and Vallès gave an example of a configuration of points admitting an unexpected curve [7, Proposition 7.3]. We recall it in Example 1.2. Actually, in [7], the authors were studying *Lefschetz properties of power ideals*, i.e., ideals generated by powers of linear forms. In [4], the authors formalize the relation between Lefschetz properties of power ideals and the existence of unexpected curves for the configuration of points dual to the linear forms that define the power ideal.

In particular, in [4], the authors gave a characterization of the existence of unexpected curves for a given set of points in terms of the *splitting type* of the dual-line arrangement. It is worth mentioning that they also relate this problem to the famous *Terao's conjecture* which claims that *freeness of a hyperplane arrangement depends only on the incidence lattice of the arrangement*. In particular, they show that, if the

splitting types depend only on the combinatorics of the arrangement, or equivalently if the existence of unexpected curves depends only on the combinatorics of the configuration of the points, then Terao's conjecture holds [4, Corollary 7.11].

In this paper, we characterize *supersolvable* line arrangements whose dual configuration of points admits unexpected curves as follows.

Theorem 3.7 *Let $\mathcal{A}$ be a supersolvable line arrangement, let d be the number of lines of $\mathcal{A}$ and let m be the maximum multiplicity of a point of intersection of the lines of $\mathcal{A}$. Then $\mathcal{A}$ admits an unexpected curve if and only if $d > 2m$; more precisely, $\mathcal{A}$ admits an unexpected curve of degree k if and only if $m \leq k \leq d - m - 1$. Moreover, when $d > 2m$, there is a unique unexpected curve of degree m .*

We also present several infinite families of line arrangements having this unexpected behavior by computing their splitting types. These families generalize examples from [4, 7]. Moreover, we want to underline that during these constructions we find examples of line arrangements which are not supersolvable (but free) satisfying the numeric condition $d > 2m$ and that do not admit unexpected curves; see, e.g., Example 4.1.

Formulation of the problem

Let $S = \mathbb{C}[x_0, x_1, x_2] = \bigoplus_{i \geq 0} S_i$ be the standard graded ring of polynomials with complex coefficients, i.e., S_i is the $\mathbb{C}$ -vector space of homogeneous polynomials of degree i . Any homogeneous ideal I inherits the grading, i.e., $I = \bigoplus_{i \geq 0} I_i$, where $I_i = I \cap S_i$.

The *fat point of multiplicity j* and *support at $P \in \mathbb{P}^2$* is the zero-dimensional scheme defined by the j th power $\wp^j$ of the ideal $\wp$ defining the point P . We denote it by jP . Observe that a homogeneous polynomial $f \in S$ belongs to $\wp^j$ if and only if all partial derivatives of f of order $j - 1$ vanish at P . This gives $\binom{j+1}{2}$ linear equations, which justifies the following definition.

Definition 1.1 Let $Z = P_1 + \cdots + P_s$ be a set of reduced points in $\mathbb{P}^2$. We say that Z admits *unexpected curves of degree $j + 1$* if, for a general point $Q \in \mathbb{P}^2$, we have that

$$\dim_{\mathbb{C}}[I(Z + jQ)]_{j+1} > \max \left\{ \dim_{\mathbb{C}}[I(Z)]_{j+1} - \binom{j+1}{2}, 0 \right\},$$

where $I(Z + jQ) = I(Z) \cap I(Q)^j$.

The general problem in this theory is the following.

Problem A *Classify all configurations of points Z that admit unexpected curves.*

If Z has general support, then it is well known that there are no unexpected curves of any degree; more generally, in [4, Corollary 6.8], the authors proved that if Z is in *linearly general position*, i.e., no three points are on a line, then it does not admit unexpected curves. The following is an example coming from [7] of a set of 9 points which admits an unexpected quartic (see also [18, Example 4.1.10]).

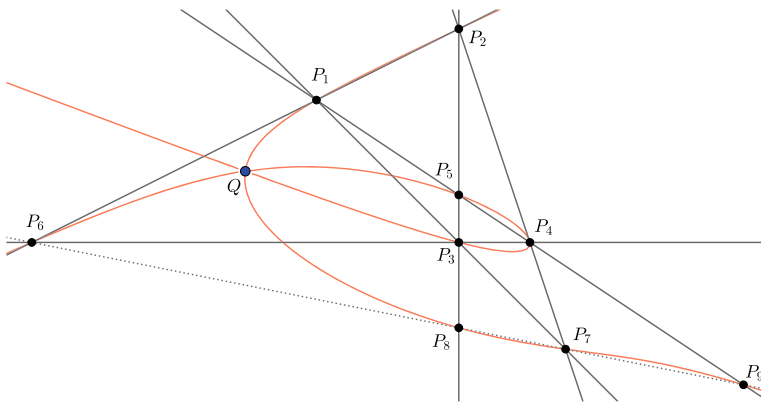


Fig. 1 The configuration unexpected quartic of Example 1.2 (Color figure online)

Example 1.2 The configuration is constructed, step by step, as follows (see Fig. 1). Consider four general points $P_1, \dots, P_4$ in the projective plane. Then, there are three pairs of lines that contain all four points. Each pair has a singular point (denoted in Fig. 1 by P_5, P_6 and P_7). Then, draw the line through two of these three points (the dotted line) and take the two points where this line intersects the pair of lines whose singular point is the third point (P_8 and P_9). This gives five additional points. If Z is the set of 9 reduced points constructed in this way, we have that $\dim_{\mathbb{C}}[I(Z)]_4 = \binom{4+2}{2} - 9 = 6$. Therefore, since the passage through a triple point $3Q$ gives $\binom{3+1}{2} = 6$ equations, we expect that, if Q is general, $\dim_{\mathbb{C}}[I(Z + 3Q)]_4 = 6 - 6 = 0$, i.e., we expect to have no quartics through Z and with an additional general triple point (denoted as the two concentric blue circles). However, there exists an unexpected quartic. See Example 3.1 for an explicit construction of this configuration in projective coordinates.

In his dissertation, Akessseh proved that if a set of points Z in the projective plane may admit an unexpected curve of degree 3 only if the ground field is of characteristic 2; see [1, Theorem 3.2.20]. Farnik et al. [9] show that this is, up to isomorphism, the only example of a configuration of points in the complex projective plane admitting an unexpected quartic.

The configuration described in the example has a very special combinatorics in relation to the B_3 arrangement (see [21, Example 1.7]). We describe it in more detail in the next section, see Example 3.1. In [4, 10], the authors connect the existence of unexpected curves for a configuration of points to the computation of the *splitting type* of the dual-line arrangement $\mathcal{A}_Z$ whose lines are defined by the linear equations having as coefficients the coordinates of the points in Z . We explain this connection in more detail later, but, in order to mention one of the main results in [4] for the reader already familiar with these combinatorial concepts, a necessary condition for a set of points Z to admit an unexpected curve in degree $j + 1$ is that $a_Z \leq j \leq b_Z - 2$, where (a_Z, b_Z) is the splitting type of $\mathcal{A}_Z$ [4, Theorem 1.5].

In this paper, we generalize Example 1.2 to infinite families of configurations having unexpected curves. In particular, while studying the problem, we noticed that

the configuration given in the Example 1.2 is the dual configuration of points to an arrangement of lines described in a paper of Grünbaum [16] where the author explains particular families of (real) line arrangements. After some experiments with the algebra software *Macaulay2* [14] and *Singular* [6], Grünbaum's paper inspired us to find the examples we describe in this paper. The families of line arrangements that we consider here are *simplicial*, i.e., arrangements of lines where every cell is a triangle, or *near simplicial*, i.e., sometimes we also have quadrilateral cells. As nicely explained in Grünbaum's paper, these arrangements occur in the literature as examples and counterexamples in many contexts of algebraic combinatorics and its applications. In this case, we related them to a new interesting question on polynomial interpolation for plane curves.

Structure of the paper

In Sect. 2, we recall the basic notions and constructions of algebraic geometry and combinatorics that we need to analyze the problem. In Sect. 3, we consider particular families of line arrangements that give unexpected curves. In this section, reader can find the main result of this paper, which is Theorem 3.7. In Sect. 4, we provide more sporadic examples of line arrangements whose dual configurations have unexpected curves, but we could not extend these to a general class of examples.

2 Basic notions and constructions

In this section, we describe the main combinatorial objects we want to consider. For more details, we refer to the classical textbook on hyperplane arrangements by Orlik and Terao [21].

Dual-line arrangement

Given a configuration of reduced points $Z = P_1 + \cdots + P_d \subset \mathbb{P}^2$, we consider the arrangement $\mathcal{A}_Z$ of dual lines $L_1, \dots, L_d$ in the dual space $(\mathbb{P}^2)^\vee$. More precisely, if $P_i = (p_{i,0} : p_{i,1} : p_{i,2})$, for any $i = 1, \dots, d$, then we define the line $L_i := \{p_{i,0}y_0 + p_{i,1}y_1 + p_{i,2}y_2 = 0\}$, where $T = \mathbb{C}[y_0, y_1, y_2]$ is the coordinate ring of the dual plane. Moreover, if $\ell_i \in T_1$ is the linear form defining the line L_i , for any $i = 1, \dots, d$, the arrangement $\mathcal{A}_Z$ is defined by the polynomial $f_Z = \ell_1 \cdots \ell_d \in T_d$.

Remark 2.1 When we say that a line arrangement admits unexpected curves, we implicitly mean that the dual configuration of points admits unexpected curves, as defined in Definition 1.1.

Splitting type of line arrangements

Let $\mathcal{A}$ be a line arrangement of d lines and let $f_{\mathcal{A}} \in T_d$ be the polynomial of degree d defining it. We consider the map defined by the gradient $\nabla_{\mathcal{A}} = [\partial_{y_0} f_{\mathcal{A}}, \partial_{y_1} f_{\mathcal{A}}, \partial_{y_2} f_{\mathcal{A}}]$

$$\mathcal{O}_{\mathbb{P}^2}^3 \xrightarrow{\nabla_{\mathcal{A}}} \mathcal{O}_{\mathbb{P}^2}(d-1).$$

We call the kernel of such a map the *derivation bundle* of $\mathcal{A}$, i.e., the rank 2 vector bundle $\mathcal{D}_{\mathcal{A}}$ defined by

$$0 \rightarrow \mathcal{D}_{\mathcal{A}} \rightarrow \mathcal{O}_{\mathbb{P}^2}^3 \xrightarrow{\nabla_{\mathcal{A}}} \mathcal{O}_{\mathbb{P}^2}(d-1).$$

Up to a twist, the derivation bundle is isomorphic to the *syzygy bundle* of the *Jacobian ideal* of $f_{\mathcal{A}}$, i.e., the ideal $J_{\mathcal{A}} = (\partial_{y_0} f_{\mathcal{A}}, \partial_{y_1} f_{\mathcal{A}}, \partial_{y_2} f_{\mathcal{A}})$ generated by the first partial derivatives of the polynomial $f_{\mathcal{A}}$.

Definition 2.2 A line arrangement $\mathcal{A}$ is said to be **free** with *exponents*, or *splitting type*, $(a_{\mathcal{A}}, b_{\mathcal{A}})$ if $\mathcal{D}_{\mathcal{A}}$ is free, i.e., if it splits as $\mathcal{D}_{\mathcal{A}} = \mathcal{O}_{\mathbb{P}^2}(-a_{\mathcal{A}}) \oplus \mathcal{O}_{\mathbb{P}^2}(-b_{\mathcal{A}})$.

In general, due to Grothendieck [15, Theorem 2.1], we know that the restriction of the derivation bundle on any line ℓ splits as $\mathcal{D}_{\mathcal{A}}|_{\ell} = \mathcal{O}_{\mathbb{P}^2}(-a) \oplus \mathcal{O}_{\mathbb{P}^2}(-b)$. The splitting type (a, b) is constant on a Zariski open subset of the dual projective plane, i.e., it is constant on a general line. We call this the *splitting type* of $\mathcal{A}$ when the arrangement is not free. For details on these facts, we refer to [4, Appendix]. If $\mathcal{A}$ is a free line arrangement, then we have that the resolution of $S/J_{\mathcal{A}}$ has length 2. In particular, the resolution is

$$0 \rightarrow T(-(d-1) - a_{\mathcal{A}}) \oplus T(-(d-1) - b_{\mathcal{A}}) \rightarrow T(-(d-1))^3 \rightarrow T \rightarrow T/J_{\mathcal{A}} \rightarrow 0,$$

where $a_{\mathcal{A}}, b_{\mathcal{A}} \in \mathbb{N}$ satisfy $a_{\mathcal{A}} + b_{\mathcal{A}} = d - 1$.

Remark 2.3 The fact that the characteristic of the field does not divide $\deg(f_{\mathcal{A}})$ is crucial for this construction. The notion of a free line arrangement can be given more generally for any characteristic, but it is more complicated and it is not needed for the purposes of this paper. For this reason, in order to make the exposition clearer, we decided to give a definition which relies on the fact that we are in characteristic 0 and we refer to [4] for the general case.

If the line arrangement $\mathcal{A}$ is actually the dual arrangement $\mathcal{A}_Z$ of a configuration of points Z , we denote its splitting type by (a_Z, b_Z) .

Conditions for unexpected curves

Finally, we give the connection between *unexpected curves* for a configuration of points and the *splitting type of the dual-line arrangement*. These are the main results in [4] that motivated this project.

In [10], the authors associate with a set of reduced points Z a *multiplicity index* defined as

$$\min\{j \mid \dim_{\mathbb{C}}[I(Z + jQ)]_{j+1} > 0, \text{ for a general point } Q\},$$

and proved that this number can be also read from splitting type, i.e., multiplicity index is equal to $\min\{a_Z, b_Z\}$. In [4], the following characterization for configurations of points which admit unexpected curves is given. Here, another important numerical character is given by

$$t_Z := \min \left\{ i \mid \dim_{\mathbb{C}}[I(Z)]_{i+1} > \binom{i+1}{2} \right\}.$$

Theorem 2.4 [4, Theorem 1.1] *Let Z be a configuration of points in $\mathbb{P}^2$ and let $\mathcal{A}_Z$ be its dual-line arrangement with splitting type (a_Z, b_Z) , say $a_Z \leq b_Z$. Then, Z admits unexpected curves if and only if $a_Z < t_Z$. In this case, Z admits an unexpected curve of degree $j + 1$ if and only if $a_Z \leq j \leq b_Z - 2$.*

Therefore, a solution to Problem A is given by the following theorem.

Theorem 2.5 [4, Theorem 1.5] *Let Z be a configuration of points in $\mathbb{P}^2$ and let $\mathcal{A}_Z$ be its dual-line arrangement with splitting type (a_Z, b_Z) . Then, Z admits an unexpected curve of degree $j + 1$ if and only if:*

- (i) $a_Z \leq j \leq b_Z - 2$;
- (ii) $\dim_{\mathbb{C}}[I(Z)]_{t_Z} = \binom{t_Z+2}{2} - |Z|$,

where $|Z|$ denotes the cardinality of the set Z .

From these results, it is clear that there is a close connection between the definition of unexpected curve for a set of points Z and the splitting type of the dual-line arrangement. Hence, our problem translates to a question about splitting types of line arrangements. As we mentioned above, the splitting type can be computed with algebra software by finding the least j such that $[I(Z + jQ)]_{j+1} \neq 0$, for a general point Q . Unfortunately, this computation is very slow and inefficient because it requires us to consider a field $\mathbb{F}$ which contains all the coordinates of the points in Z and then, if $Q = (q_0 : q_1 : q_2)$, to work over the field extension $\mathbb{F}(q_0, q_1, q_2)$.

In the next section, we focus on special line arrangements for which we can compute the splitting type and, consequently, deduce if the dual configuration of points admits an unexpected curve of some degree or not. Our computation mostly relies on the well-known Addition–Deletion Theorem and, as far as we know, this is the only theoretical tool to compute the splitting type without doing it by direct computation.

We now recall a different version of the results in [4] which is the precise way we use the aforementioned characterization of configurations of points having unexpected curves.

Theorem 2.6 [4, Theorem 1.2] *Let $Z \subset \mathbb{P}^2$ be a finite set of points and let (a_Z, b_Z) be the splitting type of the dual-line arrangement, with $a_Z \leq b_Z$. Then, Z admits an unexpected curve if and only if*

- (i) $2a_Z + 2 < |Z|$;
- (ii) *No subset of $a_Z + 2$ (or more) of the points is collinear.*

In this case, Z has an unexpected curve of degree j if and only if $a_Z < j \leq |Z| - a_Z - 2 = b_Z - 1$.

Theorem 2.6 gives a criterion for *existence* of unexpected curves, but [4] studies also conditions for *uniqueness* of such curves.

Proposition 2.7 [4, Corollary 5.5] *Let $Z \subset \mathbb{P}^2$ be a finite set of points admitting unexpected curves and let (a_Z, b_Z) be the splitting type of the dual-line arrangement, with $a_Z \leq b_Z$. Then, Z has a unique unexpected curve C in degree $a_Z + 1$. Moreover, for any $a_Z < j \leq b_Z - 1$ the unexpected curves of degree j are precisely the curves $C + L_1 + \cdots + L_r$, with $r = j - a_Z - 1$, where the L_i 's are arbitrary lines passing through the general point at which C is singular.*

In particular, if Z admits unexpected curves, then there is a unique unexpected curve of degree $a_Z + 1$.

3 Line arrangements with expected and unexpected behavior

The configuration of points constructed in Example 1.2 can be related to an interesting *simplicial* line arrangement which appears under the name $A(9, 1)$ in the list of simplicial line arrangements given by [16] (see also [5] for an updated list of these arrangements). In this section, we explain our results obtained by analyzing this list to look for new examples of unexpected curves. Here is again the Example 1.2 and the line arrangement obtained from it.

Example 3.1 We construct a configuration of points in the projective plane as described in Example 1.2. Consider the four vertices of a square: $(1 : 1 : 1)$, $(1 : -1 : 1)$, $(-1 : 1 : 1)$ and $(-1 : -1 : 1)$ and the intersection point of the diagonals of the square, i.e., the point $(0 : 0 : 1)$, and the intersections (at infinity) of the two pairs of parallel lines corresponding to the sides of the square, i.e., the points $(1 : 0 : 0)$ and $(0 : 1 : 0)$. Then, the line at infinity meets the two diagonals in two extra points $(1 : 1 : 0)$ and $(1 : -1 : 0)$. Thus, we have a set Z of nine points whose dual-line arrangement $\mathcal{A}_Z$ is defined by the polynomial $f = xyz(x+y+z)(x-y+z)(-x+y+z)(-x-y+z)(x+y)(x-y)$ and is depicted in Fig. 2b.

Now, we look at families of line arrangements generalizing the one constructed in Example 3.1. In particular, we analyze their splitting type in order to establish for which arrangements the dual configuration of points admits unexpected curves of certain degrees.

3.1 Supersolvable arrangements

We consider now a special family of line arrangements.

Definition 3.2 A line arrangement $\mathcal{A}$ is called *supersolvable* if there exists a *modular* point, i.e., a point P such that for every point $Q \in \text{Sing}(\mathcal{A})$, the line joining P and Q is an element of $\mathcal{A}$.

We denote the *multiplicity* of a point P with respect to the arrangement $\mathcal{A}$ as

$$m(P, \mathcal{A}) := |\{\ell \in \mathcal{A} \mid P \in \ell\}|.$$

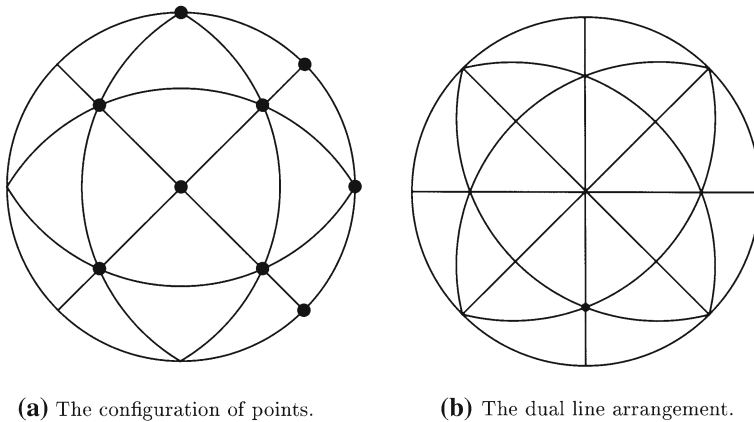


Fig. 2 The configuration of points in the projective plane and the dual-line arrangement constructed in Example 3.1. The pictures represent the projective plane, and we use the classical model of the projective plane where the line at infinity is represented by a circle on which opposite points are identified. For this reason, some straight lines are represented by circular curves

Moreover, we define $\text{Sing}_k(\mathcal{A}) := \{P \in \text{Sing}(\mathcal{A}) \mid m(P, \mathcal{A}) = k\}$ and $\text{Sing}_{\geq k}(\mathcal{A}) := \bigcup_{i \geq k} \text{Sing}_i(\mathcal{A})$.

A useful property of supersolvable line arrangements is the following.

Lemma 3.3 [2, Lemma 2.1] *Let $\mathcal{A}$ be a supersolvable line arrangement. Let $P, Q \in \text{Sing}(\mathcal{A})$ such that P is modular and Q is not. Then, $m(P, \mathcal{A}) > m(Q, \mathcal{A})$. In particular, if a point has multiplicity*

$$m(\mathcal{A}) = \max \{m(P, \mathcal{A}) \mid P \in \text{Sing}(\mathcal{A})\},$$

then it is modular.

Definition 3.4 Let $\{\ell_1, \dots, \ell_s\} \subset S_1$ be the set of the linear polynomials defining the lines of a supersolvable line arrangement $\mathcal{A}$. We say that $\mathcal{A}$ has *full rank* if $\dim_{\mathbb{C}} \text{Span}(\ell_1, \dots, \ell_s) = 3$.

Note that, by duality, the latter definition is equivalent to ask that the configuration of points dual to the line arrangement is not all contained in a line.

In this section, we want to understand when supersolvable line arrangements admit unexpected curves. We give a necessary and sufficient condition to guarantee that a supersolvable line arrangement admits no unexpected curves and then we exhibit an infinite family of cases where we have unexpected curves. This family generalizes the configuration described in Example 3.1. Our main tool is Theorem 2.6, and in order to use it, we need to compute the splitting type of supersolvable line arrangements. This is an easy application of the following well-known result which holds also in the more general setting of hyperplane arrangements.

Theorem 3.5 (Addition–Deletion Theorem; see [21, Theorem 4.51]) *Let $\mathcal{A}$ be a line arrangement in $\mathbb{P}^2$ and $\ell \in \mathcal{A}$. Let $\mathcal{A}' := \mathcal{A} \setminus \{\ell\}$. If the following conditions hold:*

- (1) $\mathcal{A}'$ is free and has splitting type (a, b) ;
 (2) $|\text{Sing}(\mathcal{A}) \cap \ell| = b + 1$ (or $a + 1$, respectively),

then $\mathcal{A}$ is free with splitting type $(a + 1, b)$ (or $(a, b + 1)$, respectively).

Now, we can compute the splitting type for supersolvable line arrangements. The following lemma is already known, as mentioned in [4, Remark 6.10]; however, for the sake of completeness and since we have not found any explicit reference, we include it here.

Lemma 3.6 *Let $\mathcal{A}$ be a supersolvable line arrangement where $d := |\mathcal{A}|$ and $m := m(\mathcal{A})$. Then, the splitting type of $\mathcal{A}$ is $(m - 1, d - m)$.*

Proof Let O be a modular point with maximal multiplicity and consider $\mathcal{A} = \mathcal{A}_0 \cup \mathcal{A}_1$, where $\mathcal{A}_0$ is the subset of lines passing through the modular point O and $\mathcal{A}_1$ is the subset of lines not passing through O . Then, $m = |\mathcal{A}_0|$ and denote $m' = |\mathcal{A}_1|$. In particular, $d = m + m'$. We proceed by induction on m' .

Let $m' = 0$. We have that $\mathcal{A} = \mathcal{A}_0$ is a central line arrangement given by m lines passing through the point O . We compute the splitting type in this case by induction on m . If $m = 2$, it is easy to check that by definition the splitting type is equal to $(1, 0)$. If $m > 2$, by the Addition–Deletion Theorem and the inductive hypothesis, we have that the splitting type of $\mathcal{A}$ is $(m - 1, 0)$.

If $m' = 1$, let $\ell \in \mathcal{A}_1$. Then,

$$\left| \bigcup_{\ell' \in \mathcal{A}_0} \ell' \cap \ell \right| = m$$

and, by the Addition–Deletion Theorem, we have that the splitting type of $\mathcal{A}$ is $(m - 1, 1)$. If $m' > 1$, let $\mathcal{A}_1 = \{\ell_1, \dots, \ell_{m'}\}$. Then, we notice that, since O is a modular point, for every pair $\ell_i, \ell_j \in \mathcal{A}_1$, the intersection $\ell_i \cap \ell_j$ lies on a line in $\mathcal{A}_0$. Therefore, for each $i = 2, \dots, m'$, if $\mathcal{A}^{(i)} = \mathcal{A}_0 \cup \{\ell_1, \dots, \ell_{i-1}\}$, we have

$$\left| \bigcup_{\ell' \in \mathcal{A}^{(i)}} \ell' \cap \ell_i \right| = m.$$

Therefore, by the Addition–Deletion Theorem and the inductive hypothesis, we conclude that the splitting type of $\mathcal{A}$ is $(m - 1, m') = (m - 1, d - m)$. $\square$

We are now ready to give a necessary and sufficient condition for supersolvable arrangements to admit unexpected curves. Here, we denote $d := |\mathcal{A}|$ and $m := m(\mathcal{A})$.

Theorem 3.7 *Let $\mathcal{A}$ be a supersolvable line arrangement, let d be the number of lines of $\mathcal{A}$ and let m be the maximum multiplicity of a point of intersection of the lines of $\mathcal{A}$. Then $\mathcal{A}$ admits an unexpected curve if and only if $d > 2m$; more precisely, $\mathcal{A}$ admits an unexpected curve of degree k if and only if $m \leq k \leq d - m - 1$. Moreover, when $d > 2m$, there is a unique unexpected curve of degree m .*

Proof Let Z be the set of points dual to $\mathcal{A}$ and let (a_Z, b_Z) be the splitting type; without loss of generality, assume that $a_Z \leq b_Z$. Now, we consider two cases.

- (1) Let $m - 1 \leq d - m$, i.e., by Lemma 3.6, $a_Z = m - 1$. Note that in this case condition (ii) of Theorem 2.6 is always satisfied: Indeed, by duality, to say that no $a_Z + 2$ points are collinear is equivalent to say that no point $P \in \text{Sing}(\mathcal{A})$ has multiplicity greater or equal to $a_Z + 2 = m + 1$. Since m is by definition the highest multiplicity of the singular points, we have that the latter condition is always true. Hence, by Theorem 2.6, we have that $\mathcal{A}$ admits unexpected curves if and only if $2(m - 1) + 2 = 2m < d$.
- (2) Let $d - m < m - 1$, i.e., by Lemma 3.6, $a_Z = d - m$ and $a_Z + 1 < m$. Now, we assume that $\mathcal{A}$ admits an unexpected curve, by condition (ii) of Theorem 2.6 we obtain that $m \leq a_Z + 1$; contradiction.

Since the condition $2m < d$ is equivalent to $m - 1 \leq d - m$, we conclude the proof. Note that the fact that, whenever $d > 2m$, $\mathcal{A}$ admits an unexpected curve of degree k for any $m \leq k \leq d - m - 1$ follows directly from Theorem 2.6. $\square$

Remark 3.8 In a recent paper, Dimca and Sticlaru [8] introduced the notion of *nearly supersolvable line arrangement*. They define a *nearly modular* point to be a point $P \in \text{Sing}(\mathcal{A})$ such that:

- (i) For any $Q \in \text{Sing}(\mathcal{A})$, with the exception of a unique point of multiplicity 2, say P' , $PQ \in \mathcal{A}$;
- (ii) $\overline{PP'} \cap \text{Sing}(\mathcal{A}) = \{P, P'\}$.

Then, a line arrangement is *nearly supersolvable* if it has a nearly modular point.

Let $\mathcal{A}$ be a nearly supersolvable line arrangement with $m = m(\mathcal{A})$ and $d = |\mathcal{A}|$. In [8, Corollary 3.2], Dimca and Sticlaru prove that the splitting type of $\mathcal{A}$ is $(d - m, m - 1)$, if $2m \geq d$, and $(\lfloor d/2 \rfloor, \lfloor d/2 \rfloor)$, if $2m < d$. By using the same idea of the proof of Theorem 3.7, it follows that nearly supersolvable arrangements do not admit unexpected curves.

In the case of supersolvable real arrangements, we have the following property.

Proposition 3.9 [2, Corollary 2.3] *Let $\mathcal{A}$ be a full rank supersolvable real arrangement. Then,*

$$|\text{Sing}_2(\mathcal{A})| + m(\mathcal{A}) \geq |\mathcal{A}|.$$

As a direct consequence, we obtain the following.

Proposition 3.10 *Let $\mathcal{A}$ be a full rank supersolvable real line arrangement such that $|\text{Sing}_2(\mathcal{A})| = \frac{d}{2}$. Then, $\mathcal{A}$ admits no unexpected curves.*

Proof By Proposition 3.9, we have $2m \geq d$. Hence, the conclusion follows by Theorem 3.7. $\square$

Remark 3.11 Examples of full rank supersolvable real arrangements with $|\text{Sing}_2(\mathcal{A})| = \frac{d}{2}$ are given by arrangements of lines dual to the so-called *Böröczky examples*.

Following [2, page 2], we see that all Böröczky examples are dual to supersolvable arrangements. In [13, Proposition 2.1], authors considered Böröczky examples because they are minimal in the sense of the famous Dirac–Motzkin conjecture which state that for a configuration of d points in real projective plane, with d large enough, there are at least $d/2$ ordinary lines, i.e., lines passing exactly through two points of the configuration; by duality, this means that the dual-line arrangement satisfies the inequality $|\text{Sing}_2(\mathcal{A})| \geq \frac{d}{2}$.

3.2 Polygonal arrangements

Now, we consider a family of arrangements included in the list of *simplicial* line arrangements given by Grünbaum in [16].

Construction

Consider a regular polygon with $N \geq 3$ edges. We consider the following arrangement:

- (1) e_i , $i = 1, \dots, N$: the lines corresponding to the sides of the N -gone;
- (2) m_i , $i = 1, \dots, N$: the lines corresponding to symmetry axes of the N -gone;
- (3) ℓ_∞ : the line at infinity.

Definition 3.12 The line arrangement $\mathcal{P}_N = \{e_1, \dots, e_N, m_1, \dots, m_N\}$ is called the N -gonal arrangement. The line arrangement $\overline{\mathcal{P}}_N = \mathcal{P}_N \cup \{\ell_\infty\}$ is called the *complete N -gonal arrangement* (Fig. 3).

Remark 3.13 Note that in the literature (see, e.g., [16]) the arrangements $\mathcal{P}_N$ are denoted by $A(2N, 1)$, while $\overline{\mathcal{P}}_N$ as $A(2N + 1, 1)$. These special configurations of lines are *simplicial arrangements*, i.e., all cells are triangles, and appear often as examples or counterexamples to various combinatorial problems.

Example 3.14 In the next figures, we describe the construction of the arrangements $\mathcal{P}_4$ and $\overline{\mathcal{P}}_4$. Note that the latter is precisely the arrangement considered in Example 3.1.

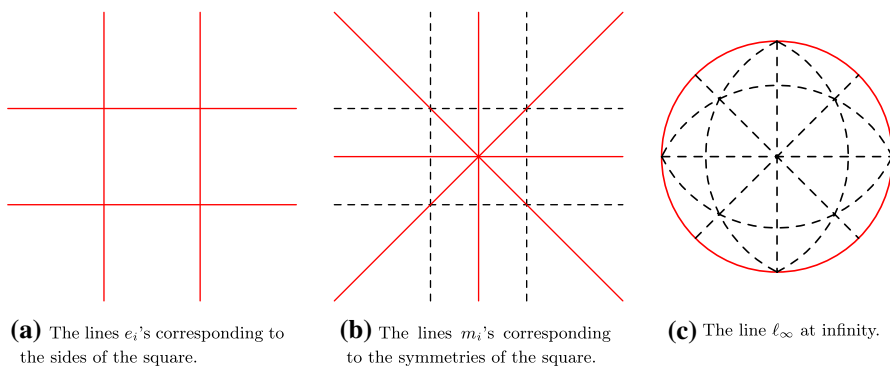


Fig. 3 Construction of $\mathcal{P}_4$ and $\overline{\mathcal{P}}_4$ (Color figure online)

Theorem 3.15 *Let $N > 2$ be an integer. Then $\mathcal{P}_N$ is always supersolvable, and $\overline{\mathcal{P}_N}$ is supersolvable if and only if N is even. Moreover, $\mathcal{P}_N$ never admits an unexpected curve, but if N is even, then $\overline{\mathcal{P}_N}$ admits a unique unexpected curve and its degree is N .*

Proof The line arrangements $\mathcal{P}_N$ is always supersolvable because all the singular points lie on a symmetry line; hence, the barycenter is a modular point. In $\overline{\mathcal{P}_N}$, we are adding the line at infinity. Note that, by Lemma 3.3, if $\overline{\mathcal{P}_N}$ is still supersolvable, then the barycenter has to be a modular point. Hence:

- (1) If N is even, every line corresponding to an edge is “parallel” to some symmetry line, i.e., they meet on the line at infinity; therefore, any singular point at infinity still lies on a symmetry line and the barycenter is still a modular point;
- (2) If N is odd, the lines corresponding to the edges are not parallel to any symmetry line; therefore, the singular points obtained as intersection of the edge lines and the line at infinity are not connected to the barycenter of the polygon, which is no longer a modular point.

By construction, the number of lines is $2N$ for $\mathcal{P}_N$ and $2N + 1$ for $\overline{\mathcal{P}_N}$. Also, $m(\mathcal{P}_N) = m(\overline{\mathcal{P}_N}) = N$. Then, our claim follows directly from Theorem 3.7. Moreover, for N even, we have that the splitting type of $\overline{\mathcal{P}_N}$ is $(N - 1, N + 1)$; therefore, by Theorem 2.6 and Proposition 2.7 there is a unique unexpected curve of degree N . $\square$

Remark 3.16 Theorem 3.15 generalizes the case described in [18, Example 4.1.10] which corresponds, in our notation, to the configuration dual to $\overline{\mathcal{P}_4}$ for which we have an unexpected quartic.

Remark 3.17 Although $\overline{\mathcal{P}_N}$ is not supersolvable when $N > 2$ is odd, and thus Theorem 3.7 does not apply, computer experiments for low odd values of N show that $\overline{\mathcal{P}_N}$ has no unexpected curves.

3.3 Tic-tac-toe arrangements

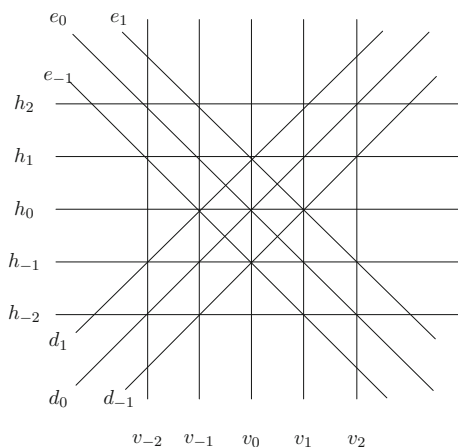
Here, we consider another family of line arrangements which generalizes [4, Example 6.14].

Construction

A tic-tac-toe arrangement of type (k, j) , denoted $\mathcal{T}_k^j$, is the arrangement defined by (Fig. 4):

1. $v_i, i = -k, \dots, k$: vertical lines $x = kz$;
2. $h_i, i = -k, \dots, k$: horizontal lines $y = kz$;
3. $d_i, i = -j, \dots, j$: the diagonals $x - y + jz = 0$;
4. $e_i, i = -j, \dots, j$: the anti-diagonals $x + y + jz = 0$.

Fig. 4 The tic-tac-toe arrangement of type $(2, 1)$



Remark 3.18 By symmetry, we may always assume that $k \geq j$. Indeed, thinking in the real projective plane, up to a 45° -rotation, we have that $\mathcal{T}_k^j$ coincides with $\mathcal{T}_j^k$. Moreover, we observe that the tic-tac-toe arrangement $\mathcal{T}_1^0$ coincides with the square arrangement $\mathcal{P}_4$ (see Fig. 3b), while, for $k > 1$, tic-tac-toe arrangements cannot be viewed as polygonal arrangements.

Similarly as above, we denote by $\overline{\mathcal{T}}_k^j$ the *complete* tic-tac-toe arrangement of type (k, j) obtained by adding also the line at infinity. In [4, Example 6.14], the authors observed that the splitting type of the complete tic-tac-toe arrangement of type $(k, 0)$ is $(2k + 1, 2k + 3)$ and they obtained the following.

Proposition 3.19 [4, Proposition 6.15] *The tic-tac-toe arrangement $\overline{\mathcal{T}}_k^0$ of type $(k, 0)$ admits a unique and irreducible unexpected curve of degree $2k + 2$.*

We may observe that $\overline{\mathcal{T}}_k^0$ is supersolvable and, in particular, free. By the Addition–Deletion Theorem, we can compute the splitting type of $\overline{\mathcal{T}}_k^1$ which, in particular, remains free. Therefore, we may inductively use the Addition–Deletion Theorem to compute the splitting type of $\overline{\mathcal{T}}_k^j$, as we show in the following.

Lemma 3.20 *Let k, j be positive integers with $k \geq j$. Then, the tic-tac-toe arrangement $\overline{\mathcal{T}}_k^j$ is free with splitting type equal to $(2k + 1 + 2j, 2k + 3 + 2j)$.*

Proof For any k and $j = 0$, we know that the claim holds by [4, Example 6.14]. We proceed now by induction on j . Assume that $\overline{\mathcal{T}}_k^j$ is free and has splitting type $(2k + 1 + 2j, 2k + 3 + 2j)$. We want to add the lines $d_{j+1}, d_{-j-1}, e_{j+1}, e_{-j-1}$ and use the Addition–Deletion Theorem four times to prove the claim for $\overline{\mathcal{T}}_k^{j+1}$. First, we need to compute the intersection between the diagonal d_{j+1} and $\overline{\mathcal{T}}_k^j$. This is:

$$|\overline{\mathcal{T}}_k^j \cap d_{j+1}| = c_1 - c_2 + c_3 = [2(2k + 1) + 1] - [2k - j] + [j + 1] = 2k + 2j + 4, \quad (1)$$

where:

- i. c_1 : the number of vertical and horizontal lines in $\overline{\mathcal{T}}_k^j$ plus the line at infinity;
- ii. c_2 : the number of points of the type $v_\alpha \cap h_\beta$ lying on d_{j+1} , i.e., the number of points of intersection in $\overline{\mathcal{T}}_k^j \cap d_{j+1}$ that are counted twice by c_1 ;
- iii. c_3 : the number of diagonals e_i 's intersecting d_{j+1} in points not of type $v_\alpha \cap h_\beta$, i.e., the remaining points in $\overline{\mathcal{T}}_k^j \cap d_{j+1}$ not yet counted by c_1 .

Then, by the Addition–Deletion Theorem, $\mathcal{T}' = \overline{\mathcal{T}}_k^j \cup \{d_{j+1}\}$ is free and has splitting type $(2k + 2 + 2j, 2k + 3 + 2j)$. Now, since $d_{-j-1} \cap d_{j+1} = d_{-j-1} \cap d_0$, we have that the cardinality of the intersection $\mathcal{T}' \cap d_{-j-1}$ is the same as counted in (1). Therefore, by the Addition–Deletion Theorem, $\mathcal{T}'' = \mathcal{T}' \cup \{d_{-j-1}\}$ is free and has splitting type $(2k + 3 + 2j, 2k + 3 + 2j)$. Similarly, since for any $\alpha, \beta, e_\alpha \cap d_\beta = v_{\alpha'} \cap h_{\beta'}$, for some α', β' , we have that also the intersections $\mathcal{T}'' \cap \{e_{j+1}\}$ and $(\mathcal{T}'' \cup \{e_{j+1}\}) \cap \{e_{-j-1}\}$ have the same cardinality as counted in (1). Again, by the Addition–Deletion Theorem, we have that the line arrangement

$$\overline{\mathcal{T}}_k^j \cup \{d_{j+1}, d_{-j-1}, e_{j+1}, e_{-j-1}\} = \overline{\mathcal{T}}_k^{j+1}$$

is free and has splitting type $(2k + 3 + 2j, 2k + 5 + 2j) = (2k + 1 + 2(j + 1), 2k + 3 + 2(j + 1))$. $\square$

Theorem 3.21 *The complete arrangement $\overline{\mathcal{T}}_k^j$ admits a unique unexpected curve of degree $2(k + j + 1)$.*

Proof Observe that $m(\overline{\mathcal{T}}_k^j) = 2k + 1$, that is, the multiplicity of one of the points at infinity, e.g., the direction of the vertical lines. Hence, in the dual configuration there are no more than $2k + 1$ collinear points. Then, we can use Lemma 3.20 and conclude by Theorem 2.6 and Proposition 2.7. $\square$

4 Other examples

In this section, we show other examples of unexpected curves arising from special line arrangements.

4.1 Adding lines to polygonal arrangements

First, we construct them by using ideas from the previous sections. We may notice that tic-tac-toe arrangements are constructed from the square arrangement $\mathcal{P}_4$ by adding lines parallel to the ones of $\mathcal{P}_4$. Hence, we try to proceed in a similar way by starting from polygonal arrangements $\mathcal{P}_N$, with N even. Unfortunately, this procedure is not successful in the sense that we can use Addition–Deletion Theorem only in a very few cases, as we are going to explain, but in general we do not know how to efficiently compute the splitting type of these line arrangements, since they are not supersolvable (hence, we cannot use Lemma 3.6) and we cannot apply Addition–Deletion Theorem.

Fig. 5 Configuration $\overline{\mathcal{P}_6}$. The line at infinity is not shown

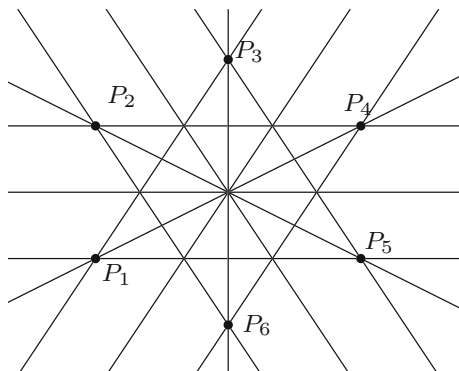
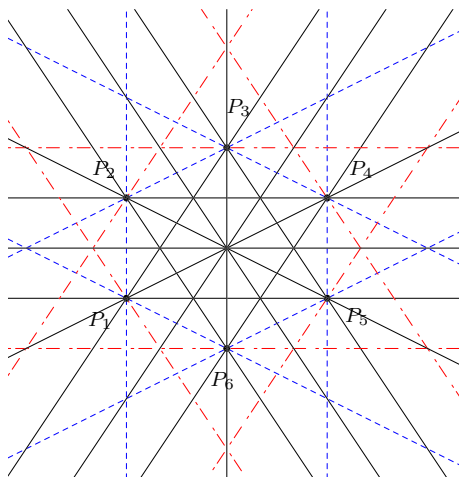


Fig. 6 Dashed blue lines $\ell_1, \dots, \ell_6$ added in the first step; dash-dotted red lines $\ell'_1, \dots, \ell'_6$ in the second (Color figure online)



Example 4.1 Consider the set of lines from the arrangement $\overline{\mathcal{P}_6}$ and take the points $P_1, \dots, P_6$ as in Fig. 5. From the proof of Theorem 3.15, we know that the splitting type for $\overline{\mathcal{P}_6}$ is $(5, 7)$. Now, we construct a series of examples for which some of the dual configuration of the points give an unexpected curve.

We add, step by step, the lines $\ell_1 := \overline{P_1 P_2}$, $\ell_2 := \overline{P_2 P_3}$, $\dots$, $\ell_6 := \overline{P_6 P_1}$ (blue dashed lines in Fig. 6). Denote $\mathcal{B}_0 := \overline{\mathcal{P}_6}$ and $\mathcal{B}_i := \mathcal{B}_{i-1} \cup \{\ell_i\}$. By Theorem 3.5, since

$$\left| \bigcup_{\ell \in \mathcal{B}_{i-1}} \ell \cap \ell_i \right| = 8, \quad \text{for } i = 1, \dots, 6,$$

the splitting types of the arrangements $\mathcal{B}_i$'s are

$$\begin{aligned} \overline{\mathcal{P}_6} &\longrightarrow \mathcal{B}_1 = \overline{\mathcal{P}_6} \cup \{\ell_1\} \longrightarrow \mathcal{B}_2 = \overline{\mathcal{P}_6} \cup \{\ell_1, \ell_2\} \\ (5, 7) &\quad (6, 7) \quad (7, 7) \\ &\longrightarrow \dots \longrightarrow \mathcal{B}_6 = \overline{\mathcal{P}_6} \cup \{\ell_1, \dots, \ell_6\}. \\ &\quad (11, 7) \end{aligned}$$

By Theorem 2.6, we have that the line arrangements $\mathcal{B}_4$, $\mathcal{B}_5$ and $\mathcal{B}_6$ admit unexpected curves of degrees 8, 9 and 10. We continue by adding lines passing through the points $P_1, \dots, P_6$, as indicated in Fig. 6 (dash-dotted red lines). Denote by ℓ'_i the new line passing through P_i , respectively, and denote the line arrangements $\mathcal{B}'_0 := \mathcal{B}_6$ and $\mathcal{B}'_i := \mathcal{B}'_{i-1} \cup \{\ell'_i\}$. Since

$$\left| \bigcup_{\ell \in \mathcal{B}'_{i-1}} \ell \cap \ell'_i \right| = 12, \quad \text{for } i = 1, \dots, 6,$$

by Theorem 3.5, the splitting types of the arrangements $\mathcal{B}'_i$'s are

$$\begin{array}{ccccccc} \mathcal{B}_6 & \longrightarrow & \mathcal{B}'_1 = \mathcal{B}_6 \cup \{\ell'_1\} & \longrightarrow & \mathcal{B}'_2 = \mathcal{B}_6 \cup \{\ell'_1, \ell'_2\} \\ (11, 7) & & (11, 8) & & (11, 9) \\ & \longrightarrow & \dots & \longrightarrow & \mathcal{B}'_6 = \mathcal{B}_6 \cup \{\ell'_1, \dots, \ell'_6\}. \\ & & & & (11, 13) \end{array}$$

We obtain three new arrangements $\mathcal{B}'_1$, $\mathcal{B}'_2$ and $\mathcal{B}'_6$ which admit unexpected curves of degree 9, 10 and 12. Moreover, we may check that the line arrangement $\mathcal{B}'_6$ shown in Fig. 6 is dual to the configuration of points given by $\text{Sing}(\overline{\mathcal{P}_6})$. This procedure of adding lines can be repeated two more times. As indicated in Fig. 7, we first add the six blue dashed lines, $m_1, \dots, m_6$ and then six red dash-dotted lines $m'_1, \dots, m'_6$. Note that the intermediate arrangements along these constructions, i.e., $\mathcal{B}'_3$, $\mathcal{B}'_4$ and $\mathcal{B}'_5$, are free arrangements which satisfy the numeric condition $d > 2m$, but that do not admit unexpected curves. Therefore, they give a counterexample to our main result (Theorem 3.7), if we drop the condition of being supersolvable. In this process, by Theorem 3.5, we get the following series of splitting types

$$\begin{array}{ccccccc} \mathcal{B}'_6 & \longrightarrow & \mathcal{B}'_6 \cup \{m_1\} & \longrightarrow & \mathcal{B}'_6 \cup \{m_1, m_2\} \\ (11, 13) & & (12, 13) & & (13, 13) \\ & \longrightarrow & \dots & \longrightarrow & \mathcal{B}''_6 := \mathcal{B}'_6 \cup \{m_1, \dots, m_6\} \\ & & & & (17, 13) \\ \mathcal{B}'_6 & \longrightarrow & \mathcal{B}''_6 \cup \{m'_1\} & \longrightarrow & \mathcal{B}''_6 \cup \{m'_1, m'_2\} \\ (17, 13) & & (17, 14) & & (17, 15) \\ & \longrightarrow & \dots & \longrightarrow & \mathcal{B}''_6 \cup \{m'_1, \dots, m'_6\}, \\ & & & & (17, 19) \end{array}$$

from which, by Theorem 2.6, we find new examples of unexpected curves.

Example 4.2 We can proceed in a similar way as in Example 4.1, but starting from configuration $\overline{\mathcal{P}_8}$. As before, we denote by $P_1, P_2, \dots, P_8$ vertices of the octagon as indicated in Fig. 8.

By Lemma 3.6, the splitting type of $\overline{\mathcal{P}_8}$ is (7, 9). We add the lines $\ell_1 := \overline{P_1 P_2}$, $\ell_2 := \overline{P_2 P_3}, \dots, \ell_8 := \overline{P_8 P_1}$ (blue dotted lines in Fig. 9). By Theorem 3.5, the splitting type

Fig. 7 Bolded points indicate the original points $P_1, \dots, P_6$; blue dotted lines $m_1, \dots, m_6$ are added in the first step; red dash-dotted lines $m'_1, \dots, m'_6$ are added in the second step (Color figure online)

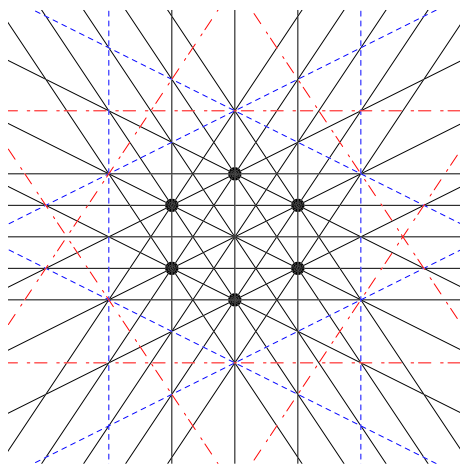
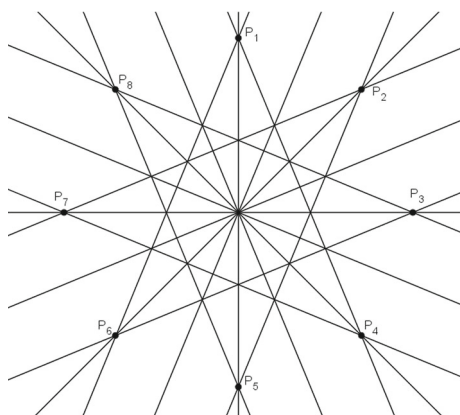


Fig. 8 Configuration $\overline{\mathcal{P}_8}$. The line at infinity is not shown



of $\overline{\mathcal{P}_8} \cup \{\ell_1, \dots, \ell_i\} = (7+i, 9)$, for all $i = 1, \dots, 8$. Thus, the existence of unexpected curves for $i \in \{4, 5, \dots, 8\}$ is guaranteed by Theorem 2.6. This line arrangement can also be extended to new arrangements which admit unexpected curves. We add eight other lines: $m_1 := \overline{P_1 P_3}$, $m_2 := \overline{P_2 P_4}$, $\dots$, $m_8 := \overline{P_8 P_2}$ (red dash-dotted lines in Fig. 9). By Theorem 3.5, the splitting type of $\overline{\mathcal{P}_8} \cup \{\ell_1, \dots, \ell_8, m_1, \dots, m_j\} = (15, 9+j)$, for all $j = 1, \dots, 8$. Thus, the existence of unexpected curves for $j \in \{1, 2, 3, 4, 8\}$ is guaranteed by Theorem 2.6.

Remark 4.3 Observe that the order of adding the lines $\ell_1, \dots, \ell_8$ in Example 4.2 (the blue dotted lines in Fig. 9) is not relevant for the computation of the splitting types; indeed, each point $Q \in \ell_i \cap \ell_j$ is also in the intersection of ℓ_i and ℓ_j with some line of $\overline{\mathcal{P}_8}$, i.e., the intersection of ℓ_i with $\overline{\mathcal{P}_8}$ is the same as the intersection of ℓ_i with any extension $\overline{\mathcal{P}_8} \cup \{\ell_j \mid \forall j \in J\}$, for some $J \subset \{1, \dots, 8\}$. Similarly, after we have added all the ℓ_i 's, the order of adding the lines $m_1, \dots, m_8$ (the red dash-dotted lines in Fig. 9 is not relevant).

Fig. 9 Blue dotted lines $\ell_1, \dots, \ell_8$ added to $\overline{\mathcal{P}_8}$ (Color figure online)

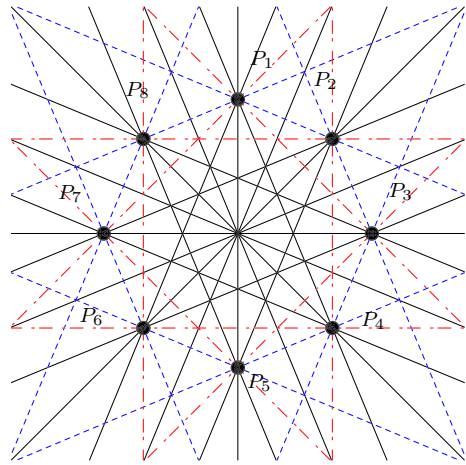


Table 1 Line arrangements dual to simplicial arrangements defined in [16] and their splitting types. The computations have been made with the algebra software *Singular*. The code used can be found as additional file to the arXiv version of the paper and includes the full computation for the $A(31, 3)$ case. The coordinates of the points of all the configurations listed in the table have been kindly provided by M. Cuntz during a private communication

$\mathcal{A}_Z$	a_Z	b_Z	$\mathcal{A}_Z$	a_Z	b_Z	$\mathcal{A}_Z$	a_Z	b_Z
$A(13, 2)$	5	7	$A(20, 5)$	8	11	$A(29, 3)$	13	15
$A(13, 3)$	5	7	$A(21, 3)$	9	11	$A(29, 4)$	13	15
$A(17, 2)$	7	9	$A(21, 4)$	9	11	$A(29, 5)$	13	15
$A(17, 4)$	7	9	$A(21, 5)$	9	11	$A(30, 3)$	13	16
$A(17, 3)$	7	9	$A(25, 2)$	11	13	$A(31, 2)$	13	17
$A(19, 1)$	7	11	$A(25, 4)$	11	13	$A(31, 3)$	13	17
$A(19, 3)$	7	11	$A(25, 7)$	11	13	$A(37, 3)$	17	19

4.2 Sporadic cases

The line arrangements presented in the previous sections are not the only ones whose dual configurations of points admit unexpected curves. There are other simplicial arrangements for which we have the same property. Table 1 shows a list of arrangements, together with their splitting type, and with their original names as in [16]. In this list, we consider line arrangements that are *dual* to the configurations $A(n, k)$ described in Grünbaum's paper. All configurations presented in Table 1, except $A(20, 5)$, are not supersolvable configurations and hence show a different type of examples than that presented in this paper so far.

Definition 4.4 Given a line arrangement $\mathcal{A}$, we define its *dual-line arrangement*, denoted by $\mathcal{A}^d$, as the line arrangement dual to the configuration of points $\text{Sing}(\mathcal{A})$.

Example 4.5 The next three figures illustrate examples of the previous definitions and notations (Figs. 10, 11, 12).

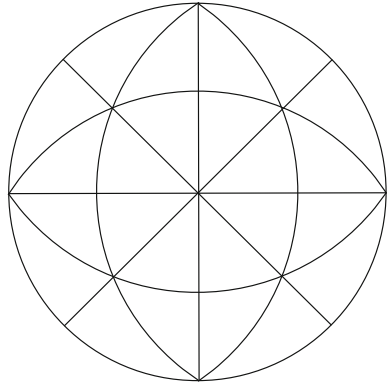
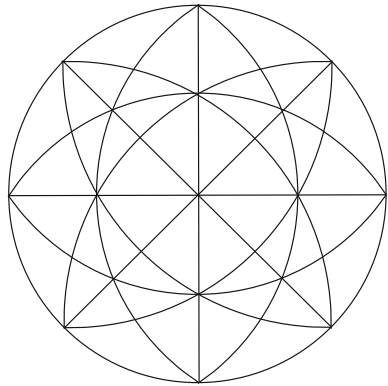
Fig. 10 Line arrangements $\overline{\mathcal{P}}_4$ **Fig. 11** Line arrangements $\overline{\mathcal{P}}_4^d$ 

Fig. 12 Line arrangement $\overline{\mathcal{P}}_4^d$ and the points in $\text{Sing}_3(\overline{\mathcal{P}}_4^d)$ (red bolded), i.e., the singular points with multiplicity at least 3. In particular, $\text{Sing}_{\geq 4}(\overline{\mathcal{P}}_4^d)$ consists only in the central point (Color figure online)

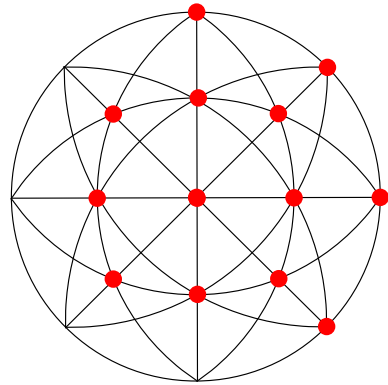


Table 2 shows a list of line arrangements such that, for some k , the configuration of points $\text{Sing}_{\geq k}$ admits unexpected curves. We also give the splitting types which speak about the degrees of unexpected curves. If the dual configuration of lines is supersolvable, we put (s) next to number $\text{Sing}_{\geq k}$ to indicate such configuration: note that, there are only four configurations in Table 2 which are of this type and that fall into the assumptions of our Theorem 3.7.

Table 2 Configurations of points defined as high-order points of some simplicial line arrangements

$\mathcal{A}$	k	$ \text{Sing}_{\geq k}(\mathcal{A}) $	$a_{\text{Sing}_{\geq k}(\mathcal{A})}$	$b_{\text{Sing}_{\geq k}(\mathcal{A})}$
$A^d(13, 2)$	4	9 (s)	3	5
$A^d(13, 2)$	3	13	5	7
$A^d(13, 2)$	1	25	11	13
$A^d(17, 2)$	4	9 (s)	3	5
$A^d(17, 4)$	4	9 (s)	3	5
$A^d(17, 4)$	3	25	11	13
$A^d(19, 3)$	4	13	5	7
$A^d(19, 3)$	3	25	11	13
$A^d(21, 3)$	4	13 (s)	5	7
$A^d(21, 3)$	3	37	17	19
$A^d(25, 2)$	4	21	9	11
$A^d(25, 4)$	4	19	7	11
$A^d(25, 7)$	4	18	7	10
$A^d(26, 3)$	4	19	7	11
$A^d(26, 4)$	4	18	7	10
$A^d(27, 2)$	4	20	8	11
$A^d(27, 3)$	4	20	8	11
$A^d(27, 4)$	4	19	7	11
$A^d(28, 4)$	4	21	9	11
$A^d(31, 3)$	4	31	13	17
$A^d(34, 2)$	5	13	5	7
$A^d(37, 3)$	4	37	17	19
$A^d(37, 3)$	6	13	5	7

4.3 Future directions

The question we considered so far (Problem A) is a special case of the following more general problem suggested by Cook et al. [4].

Problem B Let Z be a set of reduced points in $\mathbb{P}^2$ and let $\mathbb{X} = m_1 Q_1 + \cdots + m_s Q_s$ be a scheme of fat points with general support.

For which $(Z; m_1, \dots, m_s; j)$ do we have that

$$\dim_{\mathbb{C}}[I(Z + \mathbb{X})]_j > \max\{\dim_{\mathbb{C}}[I(Z)]_j - \deg(\mathbb{X}), 0\}?$$

If so, we say that Z admits unexpected curves of degree j with respect to $\mathbb{X}$.

Tables 1 and 2 show the examples of unexpected curves according to Problem A (i.e., when $\mathbb{X}$ is just a fat point), but we can also extend them to get examples of unexpected curves according to Problem B. The idea is explained in the following fact.

Proposition 4.6 *Let Z be a configuration of points with splitting type (a_Z, b_Z) with $b_Z - a_Z \geq 2$ which admits unexpected curves in the sense of Problem A. Then, for any $j \in \{0, \dots, b_Z - a_Z - 2\}$, we have that Z admits a unique unexpected curve of degree $a_Z + 1 + j$ with respect to $\mathbb{X} = (a_Z + j)P + A$, where A is a general set of reduced points of with $|A| = j$.*

Proof Let P be a general point. By Proposition 2.7, we know that, for any $j \in \{0, \dots, b_Z - a_Z - 2\}$, we have that Z admits unexpected curves of degree $a_Z + j + 1$ with respect to $(a_Z + j)P$ and, in particular, that

$$\dim_{\mathbb{C}}[I(Z + (a_Z + j)P)]_{a_Z+j+1} = j + 1.$$

Moreover, we may observe that, by definition of unexpected curves, we have

$$\max \left\{ \dim_{\mathbb{C}}[I(Z)]_{a_Z+j+1} - \binom{a_Z + j + 1}{2}, 0 \right\} \leq j. \quad (2)$$

Now, since generic simple points always impose the expected number of conditions on a linear system of curves, we have that, for any $j \in \{0, \dots, b_Z - a_Z - 2\}$,

$$\dim_{\mathbb{C}}[I(Z + (a_Z + j)P + A)]_{a_Z+j+1} = \dim_{\mathbb{C}}[I(Z + (a_Z + j)P)]_{a_Z+j+1} - j = 1;$$

and, at the same time, by (2),

$$\max \left\{ \dim_{\mathbb{C}}[I(Z)]_{a_Z+j+1} - \binom{a_Z + j + 1}{2} - j, 0 \right\} = 0.$$

Therefore, we have that Z admits a unique unexpected curve of degree $a_Z + j + 1$ with respect to $\mathbb{X} = (a_Z + j)P + A$, where A is a set of generic simple points with $|A| = j$. $\square$

For example, from Theorem 3.7 we have that supersolvable line arrangements such that $d \geq 2m + 1$ satisfy the assumptions of the latter proposition (recall that $a_Z = m - 1$ and $b_Z = d - m$). Also, some of the examples provided in Tables 1 and 2 satisfy the hypothesis of the latter proposition and give examples of unexpected curves with respect to Problem B.

Note that, by Proposition 2.7, the unexpected curves constructed in the latter proposition, whenever $j \geq 1$, are reducible. It would be interesting to construct an example of irreducible unexpected curve with respect to a scheme of fat points $\mathbb{X}$ having support in more than one point.

Finding a characterization to answer Problem B for some particular non-connected scheme $\mathbb{X}$, e.g., the union of two fat points, or constructing additional interesting examples of reducible unexpected curve besides the ones constructed in Proposition 4.6, are problems worthy of further investigation.

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